

🧠 What is the Model Context Protocol (MCP) in the context of LLMs?

In the context of Large Language Models (LLMs), the Model Context Protocol (MCP) refers to a structured format that defines how context is communicated between the user, the system, and the model. This protocol is essential for maintaining coherent interactions, managing persistent instructions, and controlling model behavior.

🔧 What does MCP include?

It includes several key components:

1. Structured Messages

- Each message has a role (`system`, `user`, `assistant`) and content.
- Example:

json

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```
{
  "role": "user",
  "content": "What is the Model Context Protocol?"
}
```

2. System Instructions

- These set the behavior of the model across the session.
- Example: "You are a helpful assistant specialized in data science and always respond in a clear and professional tone."

3. Persistent Context (Model Set Context)

- Context that remains active across interactions, such as:
 - User name
 - User goals (e.g., studying for a test or preparing for interviews)
 - Preferred tone or communication style

4. Conversation History

- The model uses a **context window** (e.g., up to 128k tokens in GPT-4o) and considers the most relevant parts of the history to generate responses.



Why is MCP Important?

- Ensures **coherence** in multi-turn conversations
- Enables **personalization** of the model's behavior
- Supports **tool usage** and memory
- Helps **reduce hallucinations** or out-of-context replies
- Facilitates **prompt engineering** and multi-message orchestration in API use



Where is MCP Used?

- In ChatGPT, when you set *custom instructions* or continue a past conversation.
- In the OpenAI API, specifically in the `chat/completions` endpoint, where developers send a sequence of messages to the model using the MCP structure.



Summary

The **Model Context Protocol (MCP)** in LLMs is a structured protocol that governs how instructions, memory, and message history are passed to the model, enabling intelligent and consistent behavior in conversations.